King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Final Exam Term 243

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If $y = \cos^4 x$, then $y'' =$

(a) $\cos^2 x \cdot (-4\cos^2 x + 3\sin^2 x)$

(b) $3\cos^2 x \cdot (\cos^2 x - 2\sin^2 x)$

(c) $-4\cos^2 x \cdot (\cos^2 x - 3\sin^2 x)$

(d) $-2\cos^2 x \cdot (2\cos^2 x - \sin^2 x)$

(e) $4\cos^2 x \cdot (\cos^2 x - 2\sin^2 x)$

2. If $y = \sqrt{\cosh(\sqrt{x})}$, then $y \cdot y' =$

(a) $2\sqrt{x}\,\sinh(\sqrt{x})$

(b) $\dfrac{\sinh(\sqrt{x})}{\sqrt{x}}$

(c) $\sinh(\sqrt{x})$

(d) $\dfrac{\sinh(\sqrt{x})}{4\sqrt{x}}$

(e) $\dfrac{\sinh(\sqrt{x})}{2\sqrt{x}}$

3. The set of all values of x at which $f(x) = \ln\left(1 - \sqrt{x-1}\right)$ is continuous is

(a) $[1, 2)$

(b) $[1, \infty)$

(c) $(1, 2)$

(d) $(-\infty, 1) \cup [2, \infty)$

(e) $[1, 4)$

4. Which one of the following statements is TRUE about the function

$$f(x) = \begin{cases} x \sin\left(\dfrac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

(a) f is not continuous at $x = 0$

(b) $f'(0) = 1$

(c) $f'(0)$ does not exist

(d) f is not continuous at $x = 1$

(e) $f'(0) = 0$

5. In an electrical circuit, the voltage V, the resistance R, and the current I are related by the equation

$$V = IR.$$

If R is increasing at a rate of 2 ohms per second and V is increasing at a rate of 3 volts per second, then the rate at which I is changing when $V = 12$ volts and $R = 4$ ohms is equal to

(a) $-\dfrac{1}{4}$ amperes/sec

(b) $-\dfrac{5}{4}$ amperes/sec

(c) $-\dfrac{3}{2}$ amperes/sec

(d) $-\dfrac{1}{2}$ amperes/sec

(e) $-\dfrac{3}{4}$ amperes/sec

6. If M and m are, respectively, the absolute maximum value and absolute minimum value of

$$f(x) = x^3 + 6x^2$$

on the interval $[-6, 1]$, then $M + m =$

(a) 22

(b) 0

(c) 32

(d) 39

(e) -7

7. $\lim_{x\to\infty}\left(\dfrac{2^x-1}{x}\right)^{1/x} =$

(a) $\dfrac{\ln 2}{2}$

(b) $\dfrac{1}{2}$

(c) $2\ln 2$

(d) 2

(e) 1

8. $\lim_{x\to 0^+} x^2(\ln x)^3 =$

(a) $-\infty$

(b) $-\dfrac{3}{4}$

(c) $-\dfrac{3}{2}$

(d) $\dfrac{3}{2}$

(e) 0

9. The graph of the function $f(x) = \dfrac{x^3}{3} - \ln x$ is

(a) decreasing on $(0,\infty)$

(b) increasing on $(0,1)$

(c) decreasing on $(1,\infty)$

(d) increasing on $(1,\infty)$

(e) increasing on $(0,\infty)$

10. The slant asymptote for the graph of $f(x) = \dfrac{2x^2-3x+2}{x-1}$ is

(a) $y = 2x+1$

(b) $y = 2x+2$

(c) $y = x+2$

(d) $y = x-2$

(e) $y = 2x-1$

11. If the graph of $f(x) = x^3 + 3kx^2 + 1$ has a local minimum at $x = 10$, then $k =$

(a) -5

(b) 5

(c) -3

(d) $\dfrac{3}{2}$

(e) $-\dfrac{3}{2}$

12. If

$$f(x) = \begin{cases} bx^2 & x \leq 1 \\ ax - 2 & x > 1 \end{cases}$$

is differentiable at $x = 1$, then $ab =$

(a) 3

(b) 6

(c) 8

(d) -1

(e) -4

13. An equation of the tangent line to the curve $y = \dfrac{1}{x}$ at the point $\left(2, \dfrac{1}{2}\right)$ is

(a) $8x + 2y = 17$

(b) $x + 4y = 1$

(c) $x + 4y = 4$

(d) $3x + 4y = 8$

(e) $x - 4y = 0$

14. The tangent line approximation of $f(x) = \dfrac{\pi}{4} - \arctan(x - 2)$ at the point $(3, 0)$ is given by

(a) $y = \dfrac{x - 3}{2}$

(b) $y = \dfrac{3 - x}{2}$

(c) $y = 1 - \dfrac{x}{3}$

(d) $y = 2x - 6$

(e) $y = 3 - x$

15. If $f(x) = \sqrt{\dfrac{x+1}{x-1}}$, then $f'(3) =$

(a) $-\dfrac{\sqrt{2}}{8}$

(b) $\sqrt{2}$

(c) $\dfrac{\sqrt{2}}{4}$

(d) $-\dfrac{\sqrt{2}}{6}$

(e) $-\dfrac{\sqrt{2}}{2}$

16. If (a, b), where $a > 0$, is the point on the curve $y = x^2$ that is closest to the point $(0, 4)$, then $ab =$

(a) $\dfrac{\sqrt{14}}{4}$

(b) $\dfrac{7\sqrt{14}}{4}$

(c) $\dfrac{7\sqrt{14}}{2}$

(d) $\dfrac{\sqrt{14}}{8}$

(e) $\dfrac{\sqrt{14}}{2}$

17. The absolute minimum value of $f(x) = \ln(x^2 + 2x + 3)$ is equal to

(a) $\ln 3$

(b) $2 \ln 2$

(c) $\ln 2$

(d) $\dfrac{1}{2} \ln 3$

(e) $\ln 6$

18. The graph of the function $f(x) = e^{-3/x}$

(a) is concave upward on $\left(\dfrac{3}{2}, \infty\right)$

(b) is concave downward on $\left(0, \dfrac{3}{2}\right)$

(c) is concave upward on $(0, \infty)$

(d) has one inflection point

(e) has two inflection points

19. If $\sinh x = -\frac{2}{3}$, then $\coth x =$

(a) $\frac{\sqrt{13}}{2}$

(b) $2\sqrt{3}$

(c) $-\frac{\sqrt{13}}{3}$

(d) $-\frac{\sqrt{13}}{4}$

(e) $-\frac{\sqrt{13}}{2}$

20. If $f'(x) = \sqrt[3]{x} + \frac{1}{2\sqrt{x}}$ and $f(1) = \frac{3}{4}$, then $f(4) =$

(a) 4

(b) $2\sqrt[3]{4} - 1$

(c) $3\sqrt[3]{4} + 1$

(d) $3\sqrt[3]{4} + 2$

(e) $\sqrt[3]{4} + 1$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$y' = -4\cos^3 x \sin x$, then product rule: $y'' = -4\cos^2 x(\cos^2 x - 3\sin^2 x)$.
2	**(d)**	$y' = \frac{1}{2y}\sinh(\sqrt{x})\frac{1}{2\sqrt{x}}$, so $yy' = \frac{\sinh\sqrt{x}}{4\sqrt{x}}$.
3	**(a)**	Need $x \geq 1$ and $\sqrt{x-1} < 1$: continuous on $[1, 2)$.
4	**(c)**	$\frac{f(h)-f(0)}{h} = \sin\frac{1}{h}$ has no limit at 0; $f'(0)$ does not exist.
5	**(e)**	$I = 3$; $3 = 3(2) + 4\,dI/dt$ gives $dI/dt = -\frac{3}{4}$.
6	**(c)**	Values $0, 32, 0, 7$ at $-6, -4, 0, 1$: $M = 32$, $m = 0$, so $M + m = 32$.
7	**(d)**	$\ln y \to \ln 2$ by L'Hopital (twice), so the limit is 2.
8	**(e)**	Write as $\frac{(\ln x)^3}{x^{-2}}$, apply L'Hopital three times: limit 0.
9	**(d)**	$f' = \frac{x^3-1}{x}$: negative on $(0, 1)$, positive on $(1, \infty)$.
10	**(e)**	Long division: $f(x) = 2x - 1 + \frac{1}{x-1}$, so $y = 2x - 1$.
11	**(a)**	$f'(10) = 0$ gives $k = -5$; $f''(10) = 30 > 0$ confirms a minimum.
12	**(c)**	Continuity $b = a - 2$, derivative $2b = a$: $b = 2$, $a = 4$, $ab = 8$.
13	**(c)**	$y' = -1/x^2$, slope $-\frac{1}{4}$ at $x = 2$: $x + 4y = 4$.
14	**(b)**	$f(3) = 0$, $f'(3) = -\frac{1}{2}$: $y = \frac{3-x}{2}$.
15	**(a)**	Chain and quotient rules: $f'(3) = \frac{1}{\sqrt{2}} \cdot (-\frac{1}{4}) = -\frac{\sqrt{2}}{8}$.
16	**(b)**	Minimize $x^4 - 7x^2 + 16$: $a = \sqrt{7/2}$, $b = \frac{7}{2}$, $ab = \frac{7\sqrt{14}}{4}$.
17	**(c)**	$f' = 0$ only at $x = -1$ ($-$ to $+$): minimum $f(-1) = \ln 2$.
18	**(d)**	$f'' = \frac{3e^{-3/x}(3-2x)}{x^4}$ changes sign only at $x = \frac{3}{2}$: one inflection point.
19	**(e)**	$\cosh x = \frac{\sqrt{13}}{3}$, so $\coth x = \frac{\sqrt{13}/3}{-2/3} = -\frac{\sqrt{13}}{2}$.
20	**(c)**	$f = \frac{3}{4}x^{4/3} + \sqrt{x} - 1$, so $f(4) = 3\sqrt[3]{4} + 1$.